

Smart Factory Production Performance & OEE Improvement

Consulting Executive Summary - Day 10



Engagement objective

Identify the principal production losses, distinguish reliability problems from throughput constraints, prioritize improvements, and quantify a simulated business case for management decision-making.

Client scenario	German industrial component manufacturer (fictional)
Factory scope	5-station serial production line, 3 shifts/day
Observation period	30 simulated working days
Data policy	100% synthetic; no confidential company data

Prepared as a portfolio case for manufacturing consulting, Industry 4.0, smart factory, automation consulting, and digital manufacturing roles.

Executive Summary

NexaMotion Components GmbH operates a simulated five-station serial production line covering CNC machining, robotic handling, quality inspection, assembly/testing, and packaging. The 30-day analysis was designed to identify where production performance is lost, which problems actually constrain finished-goods output, and which improvement initiatives create the strongest business value.

OEE 88.30% Plant equipment OEE	AVAILABILITY 96.39% Runtime / planned time	PERFORMANCE 93.21% Primary plant-level loss	QUALITY 98.28% Good / total parts
MTBF 9.74 h Functional failures	MTTR 12.45 min Average repair time	OUTPUT ATTAINMENT 97.09% Finished output vs target	ST04 CONSTRAINT 97.78% Share of shifts

Executive conclusion

Assembly & Testing (ST04) is the dominant throughput constraint, while Robot Handling (ST02) is the largest reliability/downtime problem and Quality Inspection (ST03) is the main quality-loss area. The recommended sequence is to release ST04 capacity first, then improve ST02 reliability to protect the newly available capacity.

Three distinct operational problems

Area	Loss mechanism	Evidence
ST02 - Robot Handling	Availability / reliability	Highest downtime: 1,993.35 min; availability 95.08%.
ST03 - Quality Inspection	Quality	Lowest quality: 96.81%; rejection rate about 3.19%.
ST04 - Assembly & Testing	Performance / throughput	Lowest OEE 85.46%; constraint in 97.78% of shifts.

Performance Baseline & Loss Diagnosis

The plant-level OEE of 88.30% appears strong in isolation, but decomposition by station shows that losses are not uniform. The lowest availability, lowest quality, and lowest performance occur at different stations, which means a single improvement program would not address the full problem.

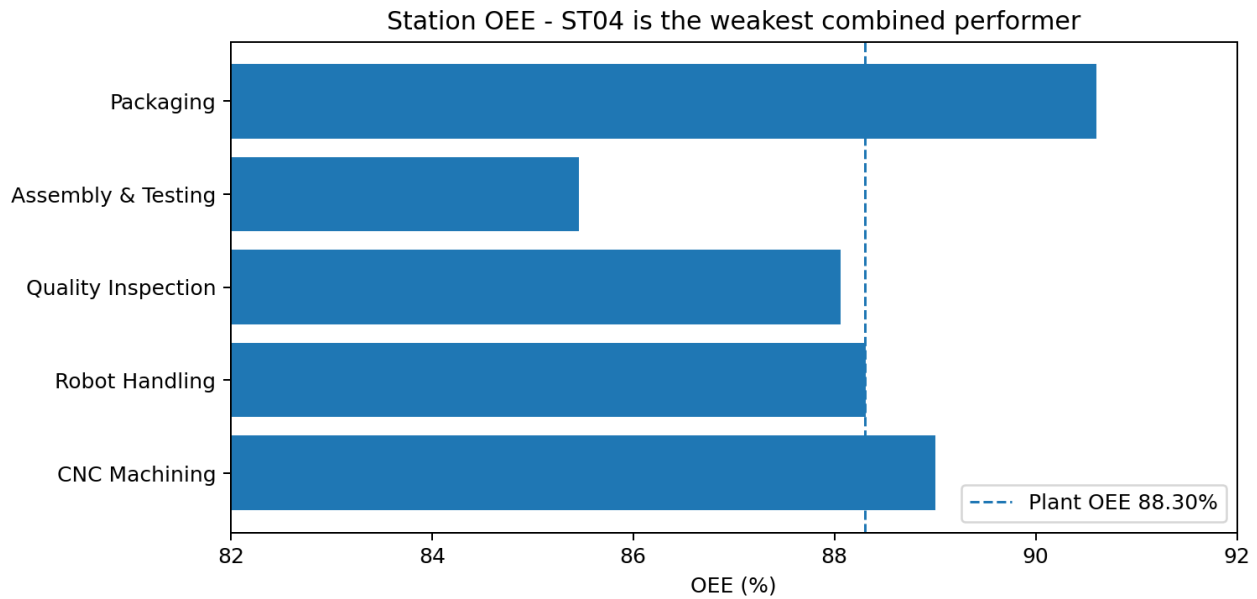


Figure 1. Station OEE comparison. ST04 has the lowest combined equipment effectiveness.

ID	Station	Availability	Performance	Quality	OEE
ST01	CNC Machining	96.69%	93.74%	98.19%	89.00%
ST02	Robot Handling	95.08%	93.58%	99.25%	88.31%
ST03	Quality Inspection	96.80%	93.96%	96.81%	88.06%
ST04	Assembly & Testing	95.58%	91.48%	97.73%	85.46%
ST05	Packaging	97.79%	93.29%	99.31%	90.60%

Root-Cause Findings

PRIMARY CASE

ST04 Assembly & Testing

- Constraint in 97.78% of shifts.
- Lowest station OEE: 85.46%.
- Lowest station performance: 91.48%.
- Slowest shift: C at 80.58 sec mean cycle time.
- Slowest product: ACT-300 at 82.23 sec mean cycle time.
- Top unplanned ST04 loss: Functional Test Failure, 439.60 min.

SECONDARY CASE

ST02 Robot Handling

- Highest station downtime: 1,993.35 min.
- Lowest station availability: 95.08%.
- Part Presence Sensor: 49 events.
- Sensor downtime: 635.46 min (10.59 h).
- Sensor share of ST02 unplanned downtime: 33.07%.
- Average sensor event: 12.97 min.

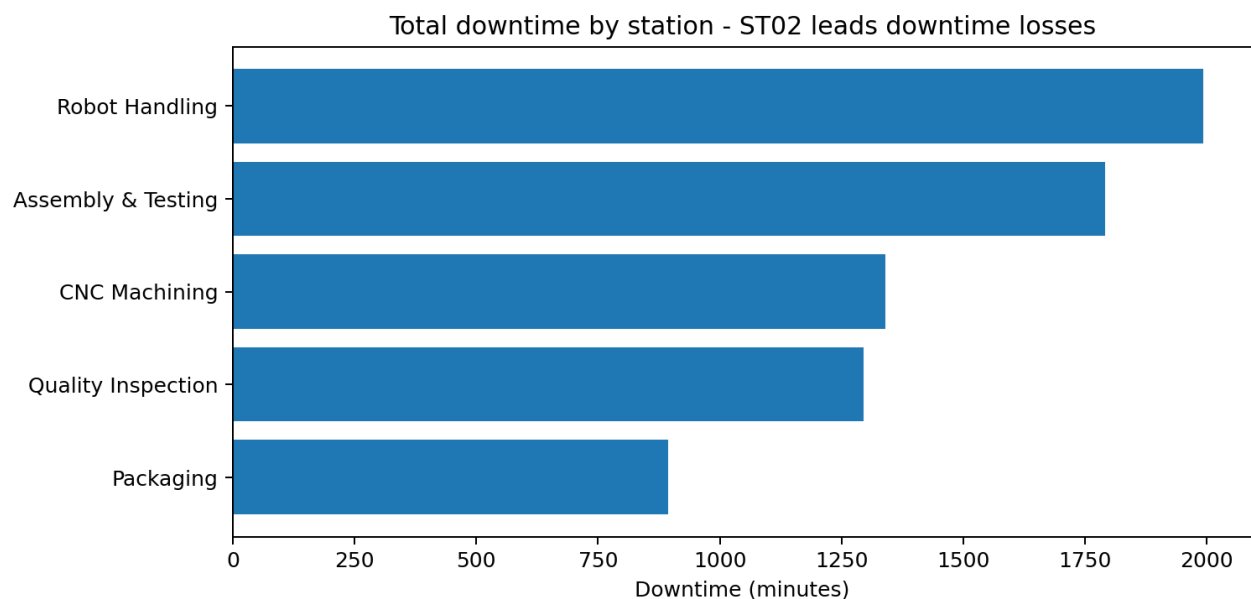


Figure 2. Downtime ranking differs from bottleneck ranking: ST02 loses the most time, but ST04 limits throughput.

Evidence vs hypothesis

What the data supports

The data supports the location and magnitude of the losses. It does not prove physical causes such as sensor misalignment, operator behavior, fixture wear, or contamination. Those remain hypotheses requiring shop-floor observation, PLC/alarm review, and process time studies.

Recommended Improvement Strategy

Phase 1 - Release ST04 capacity

- Perform a work-element time study at ST04 across all three shifts.
- Prioritize Shift C and the ACT-300 product, where the longest cycle times were observed.
- Standardize the assembly/test work sequence and remove non-value-added waiting or handling.
- Measure functional-test and fixture sub-cycle duration before committing to larger capital investment.
- Validate realized throughput improvement before moving to the next investment stage.

Phase 2 - Improve ST02 reliability

- Inspect Part Presence Sensor alignment, mounting, contamination, wiring, and product-detection conditions.
- Review PLC input history, fault codes, and alarm diagnostic quality.
- Standardize operator reset/recovery actions to reduce average recovery time.
- Implement the reliability action after ST04 capacity is released so recovered ST02 availability protects line output.

Why the sequence matters

Constraint-first principle

The ST02 sensor problem is real, but the Day 7 scenario produced zero additional finished units because ST04 remained the active bottleneck. Improving a non-constraint can improve stability without increasing plant throughput. Therefore, ST04 should be addressed first when the management objective is output growth.

Financial Case & Implementation Roadmap

YEAR-1 BENEFIT	IMPLEMENTATION	YEAR-1 ROI	PAYBACK
EUR 54.8K	EUR 15K	265.39%	3.28 mo
Simulated realized benefit	Work study + standard work	Net benefit / investment	Estimated

Business-case assumption / output	Value
Observation period	30 working days
Annual production assumption	250 working days/year
Contribution value	EUR 20 per realized additional good unit
Capacity realization factor	50%
Year-1 ramp factor	75%
Theoretical annual additional units	7,307.7
Realized Year-1 additional units	approx. 2,740
Estimated Year-1 benefit	EUR 54,807.99

Implementation roadmap

Timing	Initiative	Purpose
0-1 month	Validate ST04 losses	Time study, Shift C observation, ACT-300 work-content analysis
1-3 months	Standard work + balancing	Reduce cycle variation and non-value-added work
2-4 months	Functional-test analysis	Address 439.60 min of ST04 functional-test loss
3-6 months	Validate realized capacity	Confirm throughput and financial benefit
4-6 months	ST02 sensor reliability	Reduce recurrence and recovery time
6+ months	Evaluate larger investment	Only if ST04 remains economically significant

Final Management Conclusion

Recommended management decision

Prioritize a low-cost, evidence-led ST04 work-element and standard-work intervention before larger automation or fixture investments. Once ST04 capacity is released and the constraint is revalidated, implement the ST02 sensor reliability program to improve stability and protect the recovered capacity.

Final conclusion

Assembly & Testing is the primary production constraint. It limited the line in 97.78% of observed shifts and recorded the lowest station OEE at 85.46%. The initial recommendation is therefore a structured work-element and standard-work improvement initiative focused on ST04, particularly Shift C and ACT-300. Under simulated financial assumptions, the intervention represents approximately EUR 54.8K of Year-1 benefit against EUR 15K implementation cost, with an estimated 3.28-month payback.

Robot Handling sensor reliability is the secondary priority. The Part Presence Sensor generated 49 events and 635.46 minutes of downtime, but improving ST02 alone does not increase finished-goods output while ST04 remains the bottleneck. ST02 should therefore be treated as a follow-on reliability initiative after constraint relief.

Quality Inspection remains a secondary quality workstream. Its 96.81% quality rate indicates the largest rejection loss, but the current dataset identifies where defects are detected rather than proving where every defect is physically created. Physical root-cause validation is required before investment.

Methodology and limitation note

Item	Statement
Data source	Synthetic manufacturing records generated specifically for this portfolio project.
Analysis tools	Python, Pandas, NumPy, Excel, Power BI.
Financial values	Simulated assumptions for portfolio demonstration; not actual client savings.
Physical root causes	Hypotheses require shop-floor observation and engineering validation.
Purpose	Demonstrate manufacturing analytics, consulting reasoning, prioritization, and business-case development.